

课堂笔记整理 4.3

例 6-8

$$v = 0.1 \frac{d}{dt} (10te^{-5t}) = e^{-5t} + t(-5)e^{-5t} = e^{-5t}(1-5t)$$

$$W = \frac{1}{2} Li^2 = \frac{1}{2} \times 10.1 \times 100 t^2 e^{-10t} = 5t^2 e^{-10t} \text{ J}$$

例 6-8

$$v = 1 \times \frac{d}{dt} (90 \sin 200t) = 180 \cos 200t \text{ mV}$$

$$W = \frac{1}{2} Li^2 = \frac{1}{2} \times 1 \times (90 \times \sin 200t)^2 = 40.5 \times (\sin 200t)^2 \text{ mJ}$$

例 6-9

$$i = \frac{1}{5} \int_0^t 30t^4 dt + 0 = 6 \times \frac{t^5}{5} = 1.2t^5 \text{ A}$$

$$p = vi = 60t^5$$

$$\text{则 } W = \int p dt = \int_0^5 60t^5 dt = 10t^6 \Big|_0^5 = 156.25 \text{ J}$$

例 6-9

$$i = \frac{1}{2} \int_0^t 10(1-t) dt + 2 = (5t - \frac{5t^2}{2}) + 2 = -18 \text{ A}$$

$$W = \frac{1}{2} Li^2 = \frac{2 \times 18^2}{2} = 324 \text{ J}$$

例 6-10

$$(a) \quad i = i_L = \frac{12}{1+5} = 2 \text{ A}, \quad v_L = 5i = 10 \text{ V}$$

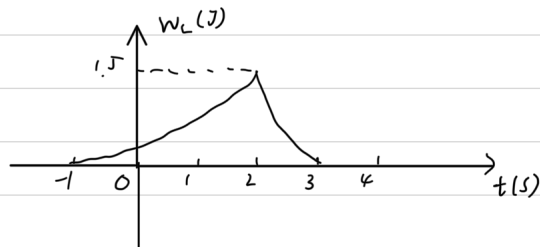
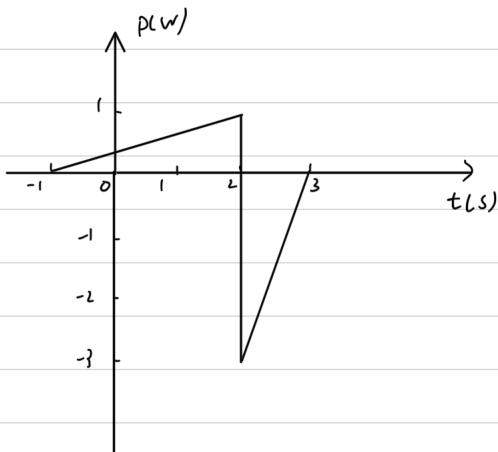
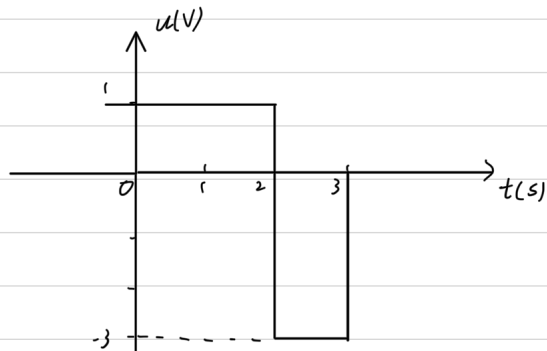
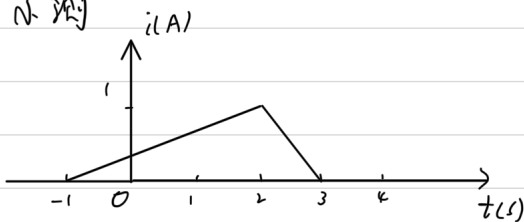
$$(b) \quad W_L = \frac{1}{2} C v_L^2 = \frac{1}{2} \times 1 \times 10^2 = 50 \text{ J}, \quad W_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \times 2 \times 2^2 = 4 \text{ J}$$

例 6-10

$$v_L = \frac{1}{3} \times 60 = 15 \text{ V}, \quad i = \frac{60}{8} = 7.5 \text{ A}, \quad W_L = \frac{1}{2} L i^2 = 3 \times 7.5 \times 7.5 = 168.75 \text{ J}$$

$$W_C = \frac{1}{2} C v^2 = \frac{1}{2} \times 4 \times 15^2 = 450 \text{ J}$$

例 6-10



例 6-11

$$7 \parallel 42 = 6H, \quad L_{eq} = 4 + 6 + 0 = 10H$$

练 6-11

$$40 \parallel 20 = 60 \quad 60 \parallel 30 = \frac{1800}{90} = 20, \quad 120 \parallel 40 = \frac{4800}{160} = 30$$

$$50 \parallel 50 = 25 \quad \text{即等效电感 } L_{eq} \text{ 为 } 25mH$$

例 6-12

$$(a) \quad i_L(0) = i(0) - i_L(0) = 4 - (-1) = 5mA,$$

$$(b) \quad L_{eq} = 2 + 4 \parallel 12 = 5H,$$

$$v_L(t) = L_{eq} \frac{di}{dt} = 5 \times 4 \times (-1) \times (-10) \times e^{-10t} mV = 200e^{-10t} mV$$

$$v_L(t) = 2 \frac{di}{dt} = 2 \times (-4) \times (-10) e^{-10t} mV = 80e^{-10t} mV$$

$$v_L(t) = v_L(t) - v_L(t) = 120e^{-10t} mV$$

$$(c) \quad i_L(t) = \frac{1}{4} \int_0^t v_L dt + i_L(0) = 8 - 3e^{-10t} mA$$

$$i_L(t) = \frac{1}{12} \int_0^t v_L dt + i_L(0) = -e^{-10t} mA$$

练 6-12

$$(a) \quad i_L(0) = i(0) - i_L(0) = 7 - 3 = 4A$$

$$(b) \quad i_L(t) = i_L(0) + \frac{1}{5} \int_0^t (-36e^{-2t}) dt = 6e^{-2t} - 2$$

$$(c) \quad v_L(t) = 6 \times 3 \times (-2) e^{-2t} = -36e^{-2t} V$$

$$v_L(t) = 8 \frac{di}{dt} = 8 \times 9 \times (-2) e^{-2t} = -144e^{-2t} V$$

$$v_L(t) = 10 \times 9 \times (-2) e^{-2t} = -180e^{-2t} V$$